# D7 - Normalization and Physical Design Audit

Table/FD rows describe the original D2 relations; R denotes their columns. V2 revision rows show the requirement-driven additions. No extra FDs are inferred from seed patterns.

| **Table** | `Member(member_id, name, email)` |
|---|---|
| Business rules | A member ID identifies one profile; each *email* belongs to at most one member (UNIQUE NOT NULL). |
| FDs | *member_id* → *name, email;*<br>*email* → *member_id* |
| Candidate keys | {*member_id*}; {*email*}. |
| Attribute closure | $\{member_id\}^+$ = R; $\{email\}^+$ = {*email*, *member_id*} → R. Empty-set closure is empty, so both keys are minimal. |
| 1NF | **Pass:** atomic values, no repeating groups, primary key exists. |
| 2NF | **Pass:** Both candidate keys are single attributes; no partial dependency. |
| 3NF | **Pass:** *name* depends on a candidate key; no non-key determinant of *name*. |
| BCNF | **Pass:** every nontrivial FD determinant is a superkey under the stated FD basis. |
| Decomposition | Not required: no normal-form violation is established. |
| Check | No split: identity mapping is lossless; all FDs remain in their original table. |

| **Table** | `Circle(circle_id, name, contribution_amount, start_date)` |
|---|---|
| Business rules | FR1/FR5: each circle has one fixed contribution *amount* for its members and one start date; the fixed interval is missing in D2. |
| FDs | *circle_id* → *name, contribution_amount, start_date* |
| Candidate keys | {*circle_id*}. |
| Attribute closure | $\{circle_id\}^+$ = R. Empty-set closure is empty; the key is minimal. |
| V2 revision | Add interval_value, interval_unit, discount_unit, discount_share_rule. *circle_id* determines all four; keys and normal forms remain unchanged. |
| 1NF | **Pass:** atomic values, no repeating groups, primary key exists. |
| 2NF | **Pass:** The only candidate key is a single attribute; no partial dependency. |
| 3NF | **Pass:** No non-key determinant of *name*, *contribution_amount* or *start_date*. |
| BCNF | **Pass:** every nontrivial FD determinant is a superkey under the stated FD basis. |
| Decomposition | Not required: no normal-form violation is established. |
| Check | No split: identity mapping is lossless; all FDs remain in their original table. |

# D7 - Normal-form audit (continued)

| **Table** | `Membership(member_id, circle_id, joined_on)` |
|---|---|
| Business rules | A member can join multiple circles; each member/circle pair has one stored joining date. |
| FDs | *(member_id, circle_id) → joined_on* |
| Candidate keys | {*member_id*, *circle_id*}. |
| Attribute closure | {*member_id*, *circle_id*}$^+$ = R; {*member_id*}$^+$ = {*member_id*}; {*circle_id*}$^+$ = {*circle_id*}. Neither proper subset determines R. |
| 1NF | **Pass:** atomic values, no repeating groups, primary key exists. |
| 2NF | **Pass:** *joined_on* depends on the whole member/circle key, not either part. |
| 3NF | **Pass:** No non-key determinant of *joined_on*. |
| BCNF | **Pass:** every nontrivial FD determinant is a superkey under the stated FD basis. |
| Decomposition | Not required: no normal-form violation is established. |
| Check | No split: identity mapping is lossless; all FDs remain in their original table. |

| **Table** | `Cycle(cycle_id, circle_id, seq_no, status)` |
|---|---|
| Business rules | Each cycle has one circle, sequence and *status*; a sequence number is unique within its circle. |
| FDs | *cycle_id → circle_id, seq_no, status;*<br>*(circle_id, seq_no) → cycle_id* |
| Candidate keys | {*cycle_id*}; {*circle_id*, *seq_no*}. |
| Attribute closure | {*cycle_id*}$^+$ = R; {*circle_id*, *seq_no*}$^+$ → {*circle_id*, *seq_no*, *cycle_id*} → R. Empty-set closure is empty; {*circle_id*}$^+$ and {*seq_no*}$^+$ contain only themselves. |
| V2 revision | Add bidding_closes_at. *cycle_id* and (*circle_id*, *seq_no*) each determine it. Derive scheduled_on from the circle interval. |
| 1NF | **Pass:** atomic values, no repeating groups, primary key exists. |
| 2NF | **Pass:** *status* depends on *cycle_id* or the whole (*circle_id*, *seq_no*) key. |
| 3NF | **Pass:** No non-key determinant of *status*; alternate-key dependencies are valid. |
| BCNF | **Pass:** every nontrivial FD determinant is a superkey under the stated FD basis. |
| Decomposition | Not required: no normal-form violation is established. |
| Check | No split: identity mapping is lossless; all FDs remain in their original table. |

# D7 - Normal-form audit (continued)

| **Table** | `Contribution(contribution_id, member_id, cycle_id)` |
|---|---|
| Business rules | FR1/FR6: one contribution record per member/cycle pair; payment *status* is required but missing in D2. |
| FDs | *contribution_id → member_id, cycle_id;*<br>*(member_id, cycle_id) → contribution_id* |
| Candidate keys | {*contribution_id*}; {*member_id*, *cycle_id*}. |
| Attribute closure | {*contribution_id*}$^+$ = R; {*member_id*, *cycle_id*}$^+$ = R. Empty-set closure is empty; {*member_id*}$^+$ and {*cycle_id*}$^+$ contain only themselves. |
| V2 revision | Add payment_status. Both candidate keys determine it. The fixed *amount* remains in Circle and is exposed per contribution by a view. |
| 1NF | **Pass:** atomic values, no repeating groups, primary key exists. |
| 2NF | **Pass:** All v1 attributes are prime. In v2, payment_status depends on the whole member/cycle key. |
| 3NF | **Pass:** Every nontrivial determinant is a superkey. |
| BCNF | **Pass:** every nontrivial FD determinant is a superkey under the stated FD basis. |
| Decomposition | Not required: no normal-form violation is established. |
| Check | No split: identity mapping is lossless; all FDs remain in their original table. |

| **Table** | `Bid(bid_id, member_id, cycle_id, discount_rate)` |
|---|---|
| Business rules | FR7/FR8: at most one bid per eligible member/cycle pair; each bid has one discount. Submission time is missing in D2. |
| FDs | *bid_id → member_id, cycle_id, discount_rate;*<br>*(member_id, cycle_id) → bid_id* |
| Candidate keys | {*bid_id*}; {*member_id*, *cycle_id*}. |
| Attribute closure | {*bid_id*}$^+$ = R; {*member_id*, *cycle_id*}$^+$ → {*member_id*, *cycle_id*, *bid_id*} → R. Empty-set closure is empty; {*member_id*}$^+$ and {*cycle_id*}$^+$ contain only themselves. |
| V2 revision | Add submitted_at. Both candidate keys determine it. Eligibility and deadline validation belong to the application workflow. |
| 1NF | **Pass:** atomic values, no repeating groups, primary key exists. |
| 2NF | **Pass:** *discount_rate* depends on *bid_id* or the whole member/cycle key. |
| 3NF | **Pass:** No non-key determinant of *discount_rate*. |
| BCNF | **Pass:** every nontrivial FD determinant is a superkey under the stated FD basis. |
| Decomposition | Not required: no normal-form violation is established. |
| Check | No split: identity mapping is lossless; all FDs remain in their original table. |

## D7 - Normal-form audit (continued)

| **Table** | `Payout(cycle_id, member_id, paid_at)` |
|---|---|
| Business rules | FR2: one payout per completed cycle. FR3/FR9 constrain eligible recipient selection across records, rather than adding a member-only FD. |
| FDs | *cycle_id* → *member_id, paid_at* |
| Candidate keys | {*cycle_id*}. |
| Attribute closure | {*cycle_id*}$^+$ = R. Empty-set closure is empty; the key is minimal. |
| 1NF | **Pass:** atomic values, no repeating groups, primary key exists. |
| 2NF | **Pass:** The only candidate key is a single attribute; no partial dependency. |
| 3NF | **Pass:** *member_id* does not determine *paid_at*; both depend on *cycle_id*. |
| BCNF | **Pass:** every nontrivial FD determinant is a superkey under the stated FD basis. |
| Decomposition | Not required: no normal-form violation is established. |
| Check | No split: identity mapping is lossless; all FDs remain in their original table. |

| **Table** | `DiscountShare(share_id, payout_cycle_id, member_id, amount)` |
|---|---|
| Business rules | Each share ID identifies one share; a recipient has at most one share for a given payout. |
| FDs | *share_id* → *payout_cycle_id, member_id, amount;*<br>*(payout_cycle_id, member_id)* → *share_id* |
| Candidate keys | {*share_id*}; {*payout_cycle_id*, *member_id*}. |
| Attribute closure | {*share_id*}$^+$ = R; {*payout_cycle_id*, *member_id*}$^+$ → {*payout_cycle_id*, *member_id*, *share_id*} → R. Empty-set closure is empty; each singleton of the composite key closes only to itself. |
| 1NF | **Pass:** atomic values, no repeating groups, primary key exists. |
| 2NF | **Pass:** *amount* depends on *share_id* or the whole payout/recipient key. |
| 3NF | **Pass:** No non-key determinant of *amount*; equal seed amounts do not imply an FD. |
| BCNF | **Pass:** every nontrivial FD determinant is a superkey under the stated FD basis. |
| Decomposition | Not required: no normal-form violation is established. |
| Check | No split: identity mapping is lossless; all FDs remain in their original table. |

## Revised design and physical design

Revised design: retain the eight base tables and original keys/FKs; add the attributes listed in the V2 revision rows (35 columns total). Existing candidate keys determine the new attributes. No normalization split is required; projection onto original columns reconstructs v1 and preserves its FDs. Views derive fixed contribution amounts, obligations, pot totals and cycle dates. Recipient eligibility, current-cycle/deadline checks and exactly one payout on completion are application transaction rules; this submission does not include workflow procedures.

Indexing: retain existing PK/UNIQUE and FK-supporting indexes. Add Bid(cycle_id, discount_rate, member_id) for retrieving a cycle's highest-discount candidates: WHERE cycle_id = ? ORDER BY discount_rate DESC. This index supports cycle filtering and discount ordering, with additional storage and write cost. Eligibility is checked separately before recipient selection.

Partitioning and denormalization: neither is justified by the small seed or a measured workload. Retain normalized relations and joins/views. Check the proposed index with EXPLAIN on representative data; its selection and any FK-index replacement are not guaranteed. No performance measurements are claimed.

Migration: migration_v1_to_v2.sql adds the seven columns, index and views in place; all 164 original seed rows, IDs and values are retained and checked. Missing metadata stays NULL; payment_status is unknown. Six historical bids are flagged for turn-order review, not removed. Back up first, stop writers and stop on errors: MySQL DDL implicitly commits. Preservation and portable views passed SQLite checks; MySQL execution has not been verified.